Classification of Physical Storage Media

Introduction

Modern computer systems use multiple types of storage devices. These devices differ in terms of **speed**, **cost**, **capacity**, and **reliability**. To simplify understanding, storage media are broadly classified into two main categories: **volatile storage** and **non-volatile storage**.

Classification of Storage Media

1. Volatile Storage

- Loses its contents once the power supply is switched off.
- Provides extremely **fast access speed**.
- Examples: **Cache memory**, **Main memory (RAM)**.
- Used for active programs and data during execution.

2. Non-Volatile Storage

- Retains its contents even after the power is turned off.
- Includes **secondary** and **tertiary** storage devices.
- Can also include **battery-backed RAM**.
- Examples: **Flash memory**, **HDDs**, **SSDs**, **Optical disks**, **Magnetic tapes**.

Factors Affecting Choice of Storage Media

- **Speed:** Determines how quickly data can be accessed.
- **Cost per unit:** Influences economic feasibility for large-scale storage.
- **Reliability:** Defines the likelihood of data corruption or hardware failure.

Storage Hierarchy

Storage devices are organized in a **hierarchical structure**, where faster and more expensive devices are placed closer to the CPU, while slower but cheaper devices are used for long-term storage.

1. Primary Storage

- **Fastest and most expensive** form of storage.
- **Volatile** – data is lost when powered off.
- Examples: **Cache memory**, **Main memory (RAM)**.
- Provides immediate access to CPU instructions and active data.

2. Secondary Storage

- Slower than primary storage but offers **larger capacity**.
- **Non-volatile** – retains data without power.
- Also known as **on-line storage**.
- Examples: **Hard Disk Drives (HDD)**, **Solid-State Drives (SSD)**, **Flash memory**.

3. Tertiary Storage

- **Slowest** type of storage, but extremely cost-effective for bulk data.
- **Non-volatile**.
- Mainly used for **archival and backup purposes**.
- Also referred to as **off-line storage**.
- Examples: **Magnetic tapes**, **Optical disks (CD/DVD/Blu-ray)**.

Example: Magnetic Tape Storage

- **Sequential access device** – slower than random-access storage.
- Storage capacity: **1 to 12 TB per tape**.
- A few tape drives can manage thousands of tapes in libraries.
- Jukebox systems provide storage in the order of **petabytes** (1000s of TB).
- Extremely **cost-effective** for long-term data backup and archival.

Storage Hierarchy Diagram

CPU Cache (SRAM) – Fastest, Volatile

Main Memory (DRAM/RAM) – Fast, Volatile

SSD / Flash Memory – Non-volatile, Very Fast

HDD / Magnetic Disks – Non-volatile, Slower

Optical Disks / Magnetic Tapes – Archival, Slowest

Storage Hierarchy

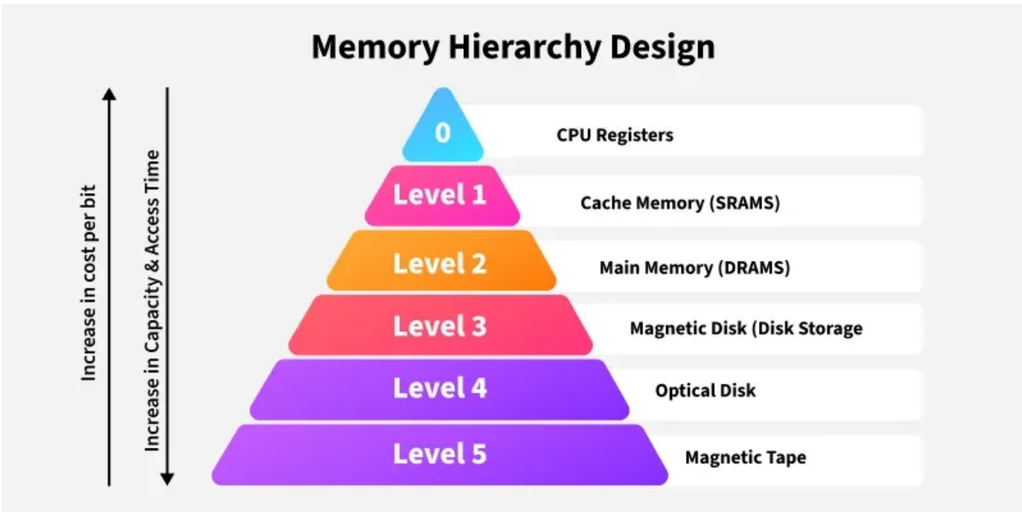

Figure 1: Storage hierarchy showing primary, secondary, and tertiary storage.